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Course Name - Artificial Intelligence

Assessment - COI/ATI

D) The goal is to measure exactly 2 gallons of water in the 4-gallon jug using only a 4-gallon and a 3-gallon jug

Let the state be represented as (x, y) , where x is the volume of water in the 4-gallon jug and y is the volume in the 3-gallon jug.

* Initial State: $(0, 0)$ * Goal State: $(2, y)$ for any y .

Step	Action	State (4G, 3G)
1	Fill 3-gallon jug	$(0, 3)$
2	Pour into 4-gallon jug	$(3, 0)$
3	Fill 3-gallon jug again	$(3, 3)$
4	Pour into 4-gallon jug until full	$(4, 2)$
5	Empty 4-gallon jug	$(0, 2)$
6	Pour remaining 2 gallons into 4-gallon jug	$(2, 0)$

$\therefore$ The 4-gallon jug contains exactly 2 gallons, which satisfies the goal.

2) The Mars Rover is an autonomous intelligent agent that explores Mars, collects samples, and sends scientific information back to Earth.

i) Percepts:

Percepts are the inputs received by the rover through sensors. They include

- * Camera images
- * Rock and soil composition
- * Surface temperature
- * Atmospheric pressure
- * Wind speed
- * Radiation level
- * GPS/position information
- * Battery status
- * Wheel traction
- * Obstacle detection

ii) Operating Environment:

Using the PEAS framework:

Environment Characteristics:

- * Partially observable
- * Sequential

- * Dynamic
- * Stochastic [Uncertain terrain]
- * Continuous
- * Single agent
- * Unknown environment

iii) Actions:

The rover can:

- * Move forward/backward
- * Turn left/right.
- * Capture images
- * Drill rocks
- * Collect samples
- * Analyse soil
- * Transmit data
- * Recharge using solar panels
- * Avoid obstacles.
- * Plan a new path.

iv) Performance Measure:

Performance is evaluated based on:

- * Number of useful samples collected
- * Accuracy of Scientific observations
- * Energy efficiency
- * Safe navigation

- * Distance explored
- * Successful communication with Earth
- * Mission completion time

V) Suitable Agent Architecture:

Model Based Goal Based Learning Agent

Why?

Because Mars is partially observable and uncertain.

The robot must:

- * Predict terrain
- * Maintain an internal map
- * Plan routes
- * Learn from previous experiences
- * Achieve mission goals independently.

Hence a Model Based Goal-Based Learning Agent is most suitable.

3) Place Eight queens so that none attack each other.

Problem Formulation:

- * Initial State: Empty chessboard.
- * State Space: Every possible arrangement of queens.
- * Operators: Place one queen in a safe position.
- * Goal Test:
 - i) 8 queens placed
 - ii) Number of two queens shared:
 - Same row • Same column
 - Same diagonal
- * Path Cost: Each placement costs 1
- * Search Strategy: Backtracking Search.

Algorithm:

- i) Start from first column.
- ii) Place queen safely.
- iii) Move to next column.
- iv) If conflict occurs:
 - Remove previous queen
 - Try next position
- v) Continue until all 8 queens are placed.

Example Solution:

One valid arrangement is

Column:	1	2	3	4	5	6	7	8
Row:	1	5	8	6	3	7	2	4

So, no queens attack each other.

AI Concepts Used:

- * State Space Research
- * Constraint Satisfaction Problem [CSP]
- * Backtracking * Depth First Search.

4) Agent Type:

This problem requires a problem Solving Agent [a specific type of Goal-Based Agent]. to find the route to the destination. However, to account for the customer's preferences [comfort, cab type], the broader system acts as a Utility-Based Agent, maximizing the user's satisfaction and efficiency.

Pseudo code:

START

Input current location

Input destination

Display available cab types

User selects cab

Check availability

IF cab available THEN

 Calculate fare

 Estimate arrival time

 Confirm booking

 Send driver's details

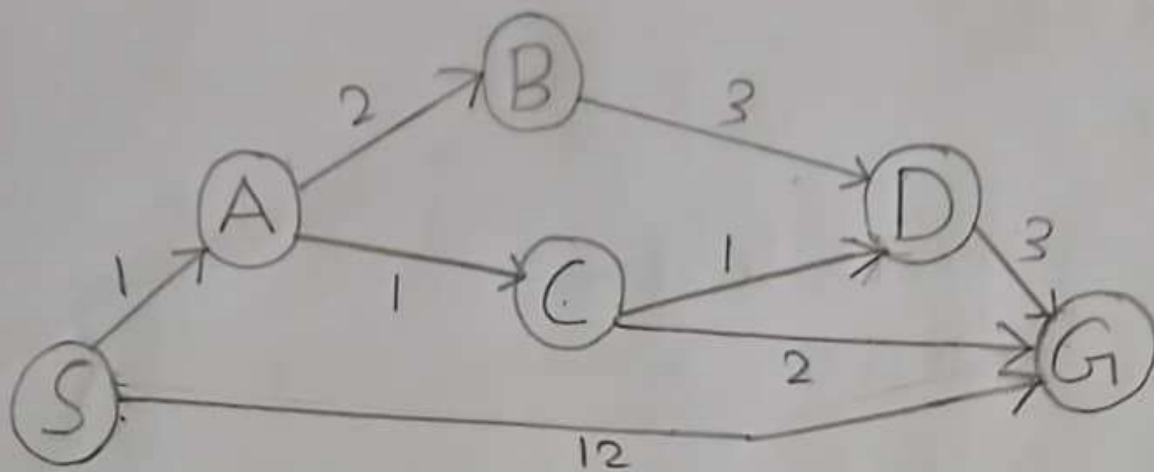
ELSE

 Show another cab option

ENDIF

Track trip

END



Based on the provided, weighted graph, the paths and costs are:

$$* S \rightarrow A = 1 \quad * B \rightarrow D = 3$$

$$* S \rightarrow G = 12 \quad * C \rightarrow D = 1$$

$$* A \rightarrow B = 2 \quad * C \rightarrow G = 2$$

$$* A \rightarrow C = 1 \quad * D \rightarrow G = 3$$

Uniform Cost Search Execution:

UCS always expands the node with the lowest cumulative path cost from the start node.

i) Initialize: Priority Queue = $\{(S, 0)\}$.

ii) Pop S (cost 0), Expand S

* Paths found: A [cost 1], G [cost 12].

* Queue : $\{(A, 1), (G, 12)\}$.

iii) Pop A (Cost 1). Expand A.

* Paths found: B (Cost $1+3=4$), C (Cost $1+1=2$).

* Queue : $\{(C, 2), (B, 4), (G, 12)\}$.

iv) Pop C (Cost 2). Expand C.

* Paths found: D (Cost $2+1=3$), G (Cost $2+2=4$).

* Queue : $\{(D, 3), (B, 4), (G, 4), (G, 12)\}$.

[We have found a cheapest path to G]
(Cost 4)

v) Pop D (Cost 3). Expand D.

* Paths Found: G (Cost $3+3=6$)

* Queue : $\{(B, 4), (G, 4), (G, 6), (G, 12)\}$

vi) Pop B (Cost 4) | G (Cost 4). [Depending on tie-breaking, if G is chosen, search ends. If B is chosen, it expands to D with Cost $4+3=7$, which is ignored. We pop G.]

vii) Pop G (Cost 4). Goal Reached!

Conclusion:

* Least Cost Path: $S \rightarrow A \rightarrow C \rightarrow G$

* Total Cost : 4